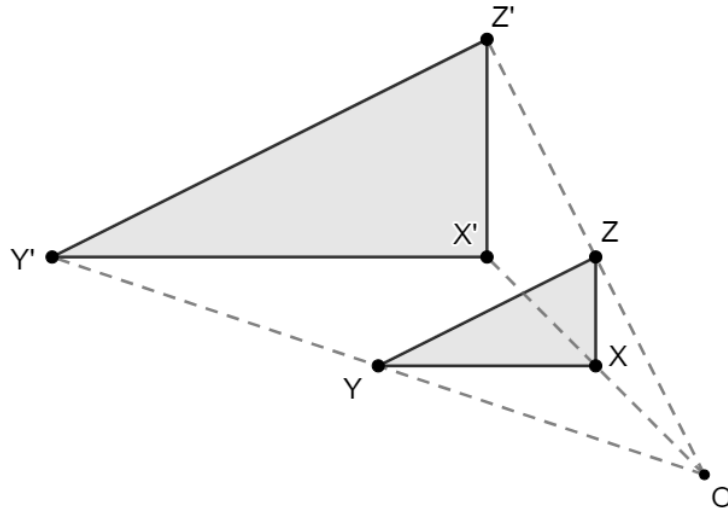


Geometry
Unit 6
Lesson 1 Practice

Name Answer Key
Date _____
Period _____

$\triangle X'Y'Z'$ was created by dilating $\triangle XYZ$. The length of $\overline{XY}$ is 2 and the length of $\overline{X'Y'}$ is 4.



1. $\triangle XYZ$ is the ☐ image ☒ preimage, $\triangle X'Y'Z'$ is the ☒ image ☐ preimage, and the center of dilation is point ☐ X ☐ X' ☒ O.

2. Determine the dilation ratio.

$$\frac{4}{2} = 2$$

3. If the length of $\overline{Y'Z'}$ is $2\sqrt{5}$, determine the length of $\overline{YZ}$.

$$\frac{2\sqrt{5}}{2} = \sqrt{5}$$

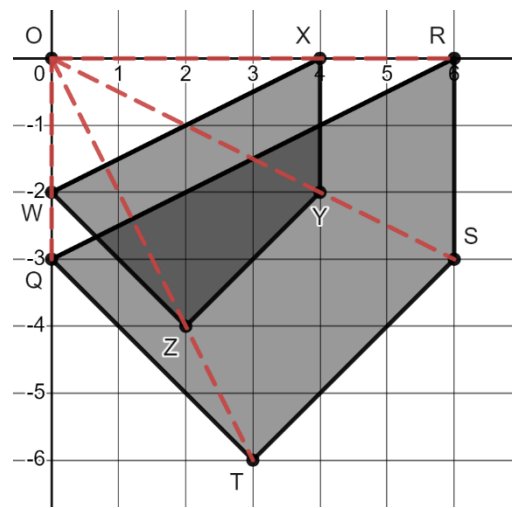
4. What is the ratio of $\frac{\overline{XZ}}{\overline{X'Z'}}$?

$$\frac{1}{2}$$

5. What is the ratio of $\frac{\overline{YY'}}{\overline{OY}}$?

$$\frac{1}{1}$$

$WXYZ$ and $QRST$ are shown on the coordinate grid, where $WXYZ$ is a dilation of $QRST$.



6. Complete the table:

Segment	Slope	Distance
$\overline{YZ}$	$\frac{2}{2} = 1$	$2\sqrt{2}$
$\overline{ST}$	$\frac{3}{3} = 1$	$3\sqrt{2}$
$\overline{WX}$	$\frac{2}{4} = \frac{1}{2}$	$2\sqrt{5}$
$\overline{QR}$	$\frac{3}{6} = \frac{1}{2}$	$3\sqrt{5}$

7. Show the dilation on the coordinate grid by drawing dotted lines from the center of dilation through each vertex.
Shown on coordinate grid.

8. Write a description of the dilation by giving the center of dilation and the scale factor.
 $QRST$ was dilated from the origin by a scale factor of $\frac{2}{3}$ to result in the image $WXYZ$.

9. What is the ratio $\frac{\overline{OX}}{\overline{OR}}$?
 $\frac{4}{6} = \frac{2}{3}$

10. What is the ratio $\frac{\overline{OX}}{\overline{XR}}$?
 $\frac{4}{2} = 2$